

Assignment No. 3

- TITLE:-

To study Alpha-Beta Tree search.

- OBJECTIVE:

To understand the Concept of Alpha-Beta Tree search and its gaming application.

- PROBLEM STATEMENT:

Implement Alpha-Beta Tree search for any game search problem.

- THEORY:

Alpha-beta pruning is a modified version of the minimax algorithm. It is an optimization technique for the minimax algorithm.

- As we have seen in the minimax search algorithm that the number of game states it has to examine is exponential in depth of the tree. Since we cannot eliminate the exponent, but we can cut it to half. Hence there is a technique by which without checking each node of the game tree we can compute the correct minimax decision and this technique is called pruning. This involves two threshold parameters Alpha and beta for future expansion so it is called alpha-beta pruning. It is also called as Alpha-Beta Algorithm.

- Alpha - beta pruning can be applied at any depth of a tree and sometimes it not only prune the tree leaves but also entire sub-tree.
- The two-parameter can be defined as:
 1. Alpha: The best (highest-value) choice we have found so far at any point along the path of Maximizer. The initial value of ^{alpha}beta is $-\infty$.
 2. Beta: The best (lowest-value) choice we have found so far at any point along path of Minimizer. The initial value of beta is $+\infty$.
- The Alpha - beta pruning to a standard minimax algorithm returns the same move as the standard algorithm does, but it removes all the nodes which are not really affecting to final decision but making algorithm slow. Hence by pruning these nodes, it makes the algorithm fast.

The main condition which required for alpha-beta pruning is:

1. $\alpha \geq \beta$

- Algorithm Steps:-

function AlphaBeta (node , depth , alpha , beta , maximizing player):

if depth == 0 or node is a terminal node:
return the heuristic value of node

if maximizingPlayer:

maxEval = $-\infty$

for each child of node:

eval = AlphaBeta (child , depth - 1 , alpha , beta , false)

maxEval = max (maxEval , eval)

alpha = max (alpha , eval)

if beta $\leq$ alpha:

break # Beta cutoff (prune the branch)

return maxEval

else:

minEval = ∞

for each child of node:

eval = AlphaBeta (child , depth - 1 , alpha , beta , true)

minEval = min (minEval , eval)

beta = min (beta , eval)

if beta $\leq$ alpha:

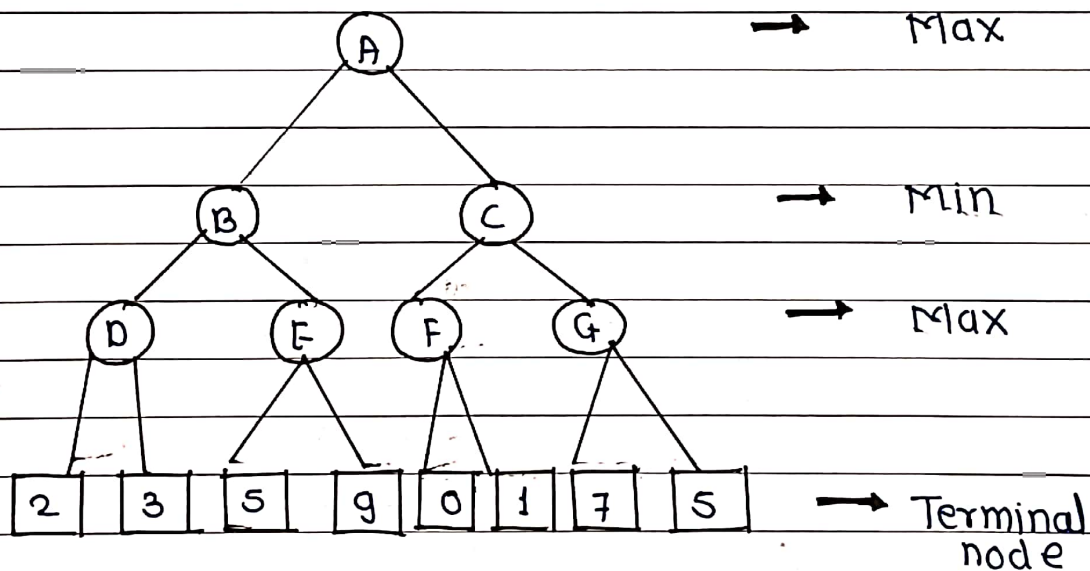
break # Alpha cutoff (prune the branch)

return minEval.

Working of Alpha-Beta Pruning:-

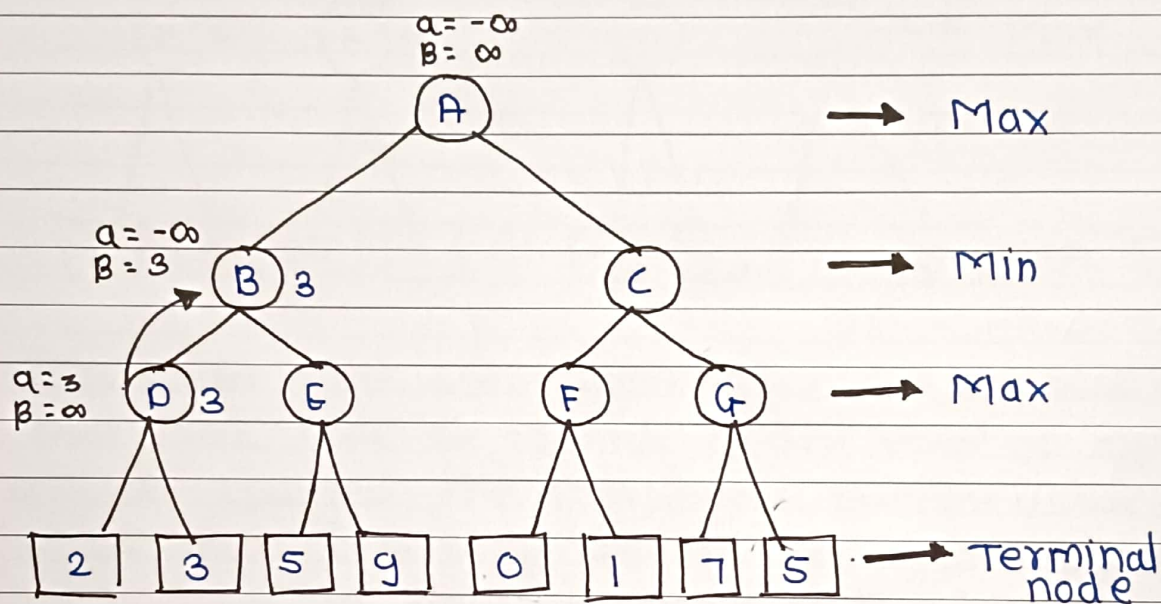
Let's take an example of two player search tree to understand the working of Alpha-beta pruning.

Step 1: At the first step the Max player will start first move from node A where $\alpha = -\infty$ and $\beta = +\infty$, these value of alpha and beta passed down to node B same where value to its child D.



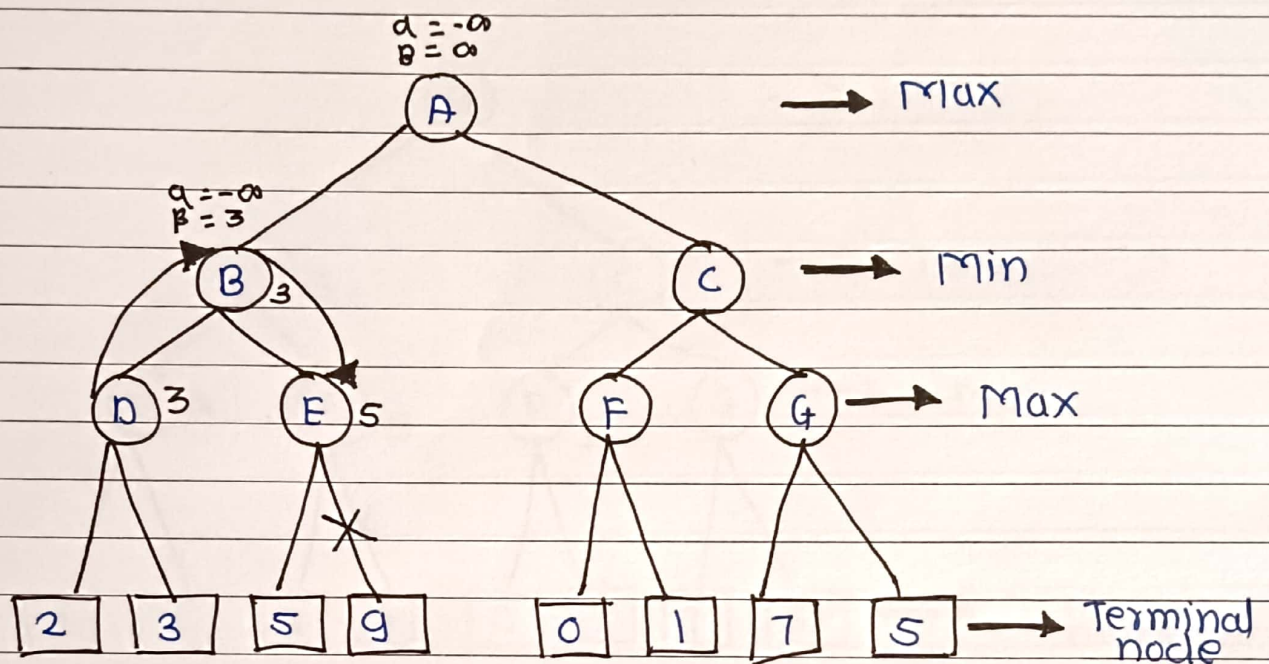
Step 2: At Node D, the value of α will be calculated as its turn for Max. The value of α is compared with firstly 2 and then 3 and the max $(2, 3) = 3$ will be the value of α at node D and node value will also 3.

Step 3: Now algorithm backtrack to node B where the value of B will change as this is a turn of Min. Now $\beta = +\infty$ will compare with the available subsequent node value i.e. $\min(\infty, 3) = 3$ hence at node B now $\alpha = -\infty$ and $\beta = 3$.



In the next step algorithm traverse the next successor of Node B which is node E and the value of $\alpha = -\infty$ and $\beta = 3$ will also be passed.

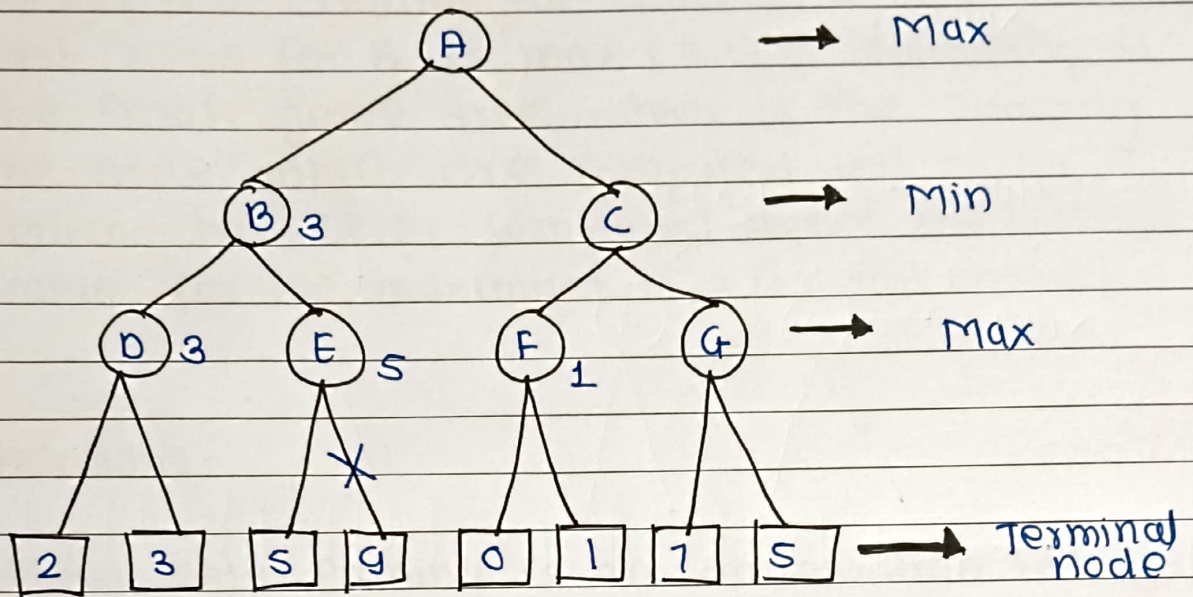
Step 4: At node E, Max will take its turn and the value of alpha will change. The current value of alpha will be compared with 5 so $\max(-\infty, 5) = 5$ hence node E $\alpha = 5$ and $\beta = 3$ where $\alpha \geq \beta$ so the right successor of E will be pruned and algorithm will not traverse it and the value at node E will be 5.



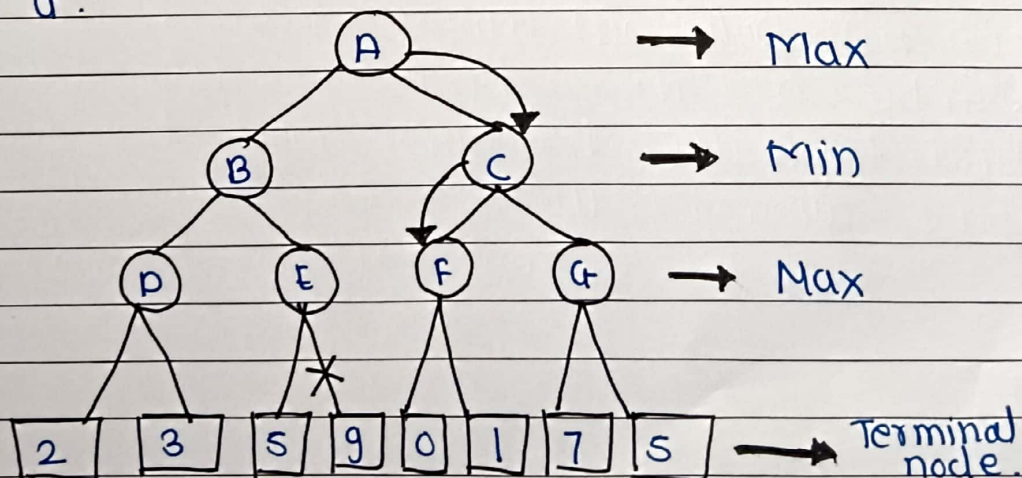
Steps 5: At next step algorithm again backtrack the tree from node B to node A the value of alpha will be changed the maximum available value is 3 as $\max(-\infty, 3) = 3$ and $\beta = +\infty$ these two values now passes to right successor of A which is Node C

At node C, $\alpha = 3$ and $\beta = +\infty$ and the same values will passed on to node F

Steps 6: At node F, again the value of α will be compared with left child which is 0 and $\max(3, 0) = 3$ and then compared with right child which is 1, and $\max(3, 1) = 3$ still α remains 3 but the node value of F will become 1.



step 7: Node f return the node value 1 to node C at at C $\alpha = 3$ and $\beta = +\infty$ here the value of beta will be changed it will compare with 1 so $\min(\infty, 1) = 1$ Now at C, $\alpha = 3$ and $\beta = 1$, and again it satisfies the condition $\alpha \geq \beta$ so the next child of C which is G will be pruned, and the algorithm will not compare the entire sub tree G.



Step 8:- C Now returns the value of 1 to A here the Best value for A is $\max(3, 1) = 3$ following is the final game tree which is showing the node which are computed and nodes which has never computed Hence the optimal value for the maximizer is 3 for this example.

Conclusion:-

Alpha Beta Pruning is an optimization technique for the minimax by pruning unnecessary branches thus speeding up the decision making process while ensuring the same optimal result.